Dr. Bindu Devi

Ph. D (MICROBIOLOGY)

Educational qualifications:

Class / Degree	Board / Institute/University		Subject(s)	
B.Sc	HPU, Shimla		Botany, Zoology, Chemistry	
M.Sc (Microbiology)	CSKHPKV, Palampur		Major : Microbiology Minor: Biochemistry	
Ph.D (Microbiology)	Dr. YSPUHF, Nauni, Solan		Major : Microbiology Minor: Biotechnology	

M.Sc THESIS TITLE: Optimization of growth conditions of indigenous *Lactobacillus* spp. and their interaction studies with food-borne pathogens.

Ph. D THESIS TITLE: Purification of important bioactive compounds synthesized by functionally proficient probiotics and their genetic characterization

Techniques:

- Microscopy, Cultivation and isolation of microbes, Maintenance and preservation of pure cultures, Gel Filtration Chromatography, Morphological and biochemical identification of microbes, DNA isolation and PCR.

***Instruments handled*:**

- Autoclave, Laminar air flow chamber, Spectrophotometer, Centrifuge, Anaerobic work station, pH meter, Soxhlet apparatus, Gel electrophoresis and PCR.

PUBLICATIONS

Book Chapters:

- Sunita Devi, **Bindu Devi** and Seema Verma. 2017. Practices in bioleaching: a review on clean and economic alternative for safe and green environment Apple Academic Press, USA.
- Nivedita Sharma, **Bindu Devi** and Moksharthi Sharma. 2021. Impressive antimicrobial potential of human milk: A blessing to nascent life. Nova Science Publishers, USA.
- **Bindu Devi**, Sunita Devi, Diksha Sood and Priyanka Arya. 2023. Microbial stabilizers in food processing. Springer, Singapore.
- Diksha Sood, Sunita Devi, **Bindu Devi** and Priyanka Arya. 2023. Microbial antioxidants in food products. Springer, Singapore.
- Nivedita Sharma, Nisha Sharma, Shakshi Sharma, Pushpinder Sharma and **Bindu Devi**. 2023. Identification, morphological, biochemical and genetic characterization of microorganisms. Academic Press.
- Dimple, Nivedita Sharma, Nisha Sharma, Neha Gautam, **Bindu Devi** and Sudarshna. 2023. Microbial Pectinase Production: A boon for various Industries. Academic Press.

Research Articles:

- Parul Sharma, Amreen, Geeta Rawat, **Bindu Devi** and Mohinder Kaur. 2024. Exploring Actinomycete diversity and growth potential in replant sites of apple orchards in Shimla District, Himachal Pradesh. International Journal of Advanced Biochemistry Research. 8(6): 218-225.
- Rialch A, Kumari S, Devi B, Kaushal N, Mal G, Banerjee PS, Sankar M, Singh B, Sharma R, Gopinath D, Jairath G, Katoch S, Prerna M, Thakur R. (2025). Unveiling the genetic diversity of bovine *Cryptosporidium* isolates in North India inferred by 18S rRNA sequences. Microbial Pathogenesis. **(Communicated)**
- **Bindu Devi** and Nivedita Sharma. 2025. Evaluation of Physicochemical, Organoleptic and Fatty-acid Composition of High Altitude Trans-Himalayan Goat Milk Fermented with Probiotic Isolate- *L. fermentum* 214 (NCDC 214). **(Communicated)**
- **Bindu Devi** and Nivedita Sharma. 2025. Isolation and purification of casein peptide and Fatty-acid profiling of Gaddi Tribe Sheep's milk of North-Western Himalayas Fermented with Probiotic Isolate- *L. fermentum* L2 (MF443393). **(Communicated)**
- **Bindu Devi** and Nivedita Sharma. 2025. Evaluation of Physicochemical, Organoleptic and Fatty-acid Composition of High Altitude Trans-Himalayan Goat Milk Fermented with Probiotic Isolate- *L. fermentum* 214 (NCDC 214). **(Communicated)**

Review Articles:

- Aryan S., **Devi**, **B.,** Kumar, A., Sinha S., and Sharma, P. (2025). Cultural Flavor: Exploring Fermented and Non-Fermented Foods and Beverages of the Jaunsar-Bawar Tribe of Uttarakhand. Plant Archives. 25: 1807-1816. DOI:10.51470/PLANTARCHIVES.2025.v25.no.1.263
- Giri, N., Tomar, G., Singh, K., Kotiyal, S., & **Devi, B**. (2025). Antibiotic Resistance. *International Journal of Zoological Investigation* 11:777-792. DOI:10.33745/ijzi.2025.v11i01.082
- Sharma P, Rawat G, Gupta P, **Devi B** and Nautiyal D. (2025). Siderophore: enhancing crop nutrition and improving stress tolerance through microbial iron Scavenging. *Asian Pacific Journal of Molecular Biology and Biotechnology*. 33:183-196. DOI:10.35118/apjmbb.2025.033.2.18

Book Editor:

- Mishra, N., Chanu, S. N., Rathore, M. S., **Devi, B.,** Patidar, A., & Saini, A. (2024). *Modern Horizons in Agricultural Sciences*. KD Publication.